Sachi Patel

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Education

Columbia University — Expected May 2028
Accelerated M.S. (4+1) in Computer Science, Machine Learning Pathway — *New York, NY*

Barnard College of Columbia University — Expected May 2027
B.A. in Computer Science & Economics | GPA: 3.83 | New Pathways Bridgewater Scholar — *New York, NY*

- **Selected Coursework:** Data Structures, Computer Architecture, Systems Programming, Natural Language Processing, Artificial Intelligence, Applied Machine Learning, Databases, Agentic Engineering, Analysis of Algorithms, Discrete Math, Linear Algebra

Skills

Languages/Frameworks: Python, Java, JavaScript, TypeScript, C, SQL, Bash, R; React, Node.js, Flask, FastAPI, Next.js
ML/AI: PyTorch, scikit-learn, Hugging Face, LangChain, Google ADK, MCP, RAG, FAISS, Pandas, NumPy
Cloud/Infra: AWS (CDK, ECS Fargate, EC2, S3, Lambda, CloudWatch), Terraform, Docker, GitLab CI/CD
Observability/Data: OpenTelemetry (OTLP), Prometheus, Grafana, PostgreSQL, BigQuery

Work Experience

Goldman Sachs — Jun. 2026 – Present
Software Engineering Intern – Data & AI Solutions Engineering — *New York, NY*

- Building multi-tenant observability platform on AWS serving 15+ teams, unifying app and infra metrics across 100s of services
- Designing metrics pipeline from OTLP ingestion to lakehouse storage; shipping Grafana dashboards/Alertmanager routing
- Developing agent skills and MCP tooling over lakehouse metrics/logs to automate error surfacing for SRE incident response

Depository Trust and Clearing Corporation — Jun. 2025 – Aug. 2025
Software Engineering Intern – Cloud Center for Enablement — *Jersey City, NJ*

- Built internal automation to detect idle EC2/EBS volumes across 25+ SYSIDs; alerts drove $350K in annualized savings
- Developed MVP cost-savings dashboard (Python, Flask, REST APIs) integrating CloudAware and AWS Cost Explorer data
- Presented cost insights to CloudOps teams and SYSID owners; drove decommissioning of 200+ idle resources org-wide

Rajiv Sethi Lab, Barnard College — Jan. 2024 – Present
Research Assistant — *New York, NY*

- Built data pipeline (Python, BeautifulSoup) scraping price data/probabilities from prediction markets/models across 41 events
- Implemented profitability test (Python/MATLAB) analyzing trading strategies; extracted insights on model vs. market accuracy
- Co-authored paper finding FiveThirtyEight beat Polymarket by 17.9% on popular vote; published in ACM CI '25 proceedings

Martina Jasova Lab, Barnard College — Jul. 2024 – Dec. 2024
Research Assistant — *New York, NY*

- Developed text extraction pipeline (PyMuPDF, concurrent.futures) to parse and chunk 12K+ pages of financial reports
- Integrated GPT-4o mini for ESG sentiment classification with batched inference, retry logic, and exponential backoff
- Computed alignment scores (cosine similarity) between 50+ banks' ESG claims and actual sustainable lending activity

Barnard Computational Science Center (CSC) — Aug. 2024 – Present
Senior Computing Fellow — *New York, NY*

- Mentored 120+ students in Python, R, and MATLAB across 3 data science courses; debugged code and guided final projects
- Led 30+ office hours sessions supporting Coding Markets, Privacy in a Networked World, and Intro to Data Science
- Designed and led 3 workshops with modular Colab exercises covering data wrangling, visualization, and scripting

Projects

Agentic Stock Prediction System — Jun. 2026 (In Progress)

- Building autonomous trading agent (Google ADK) that orchestrates tool-calling across market data APIs/technical indicators
- Integrating LLM reasoning over real-time and historical market data to generate buy/sell/hold recommendations with rationale
- Designing evaluation harness that backtests agent recommendations against historical returns to measure decision quality

RAG-Powered Codebase Assistant — Jan. 2026

- Built RAG assistant that ingests GitHub repos, parses Python code with tree-sitter, answers questions via natural language queries
- Implemented semantic search using sentence-transformers/FAISS; integrated LangChain + Ollama for conversational UI
- Engineered multi-repo indexing with chunked parsing; achieved sub-second query latency across 10K+ lines of code

HTTP Web Server — Dec. 2024

- Built HTTP/1.0 server in C using POSIX sockets; forked child processes to handle concurrent connections and serve static files
- Implemented request parsing, URI validation, directory traversal prevention, and error handling (200/301/400/404/501)

Activities & Leadership

Columbia Multicultural Business Association (Conference Lead) | Society of Women Engineers (Committee Member) | Columbia Women in Computer Science (Event Coordinator) | Grace Hopper Conference '24 & '25 | Columbia DJ Collective